# Homework #11 Solution

Consider the LP

$$\min_x \; f(x) = x_1 + x_3$$
$$\text{s.t.} \quad x_1 + x_2 + x_3 = 1$$
$$x \geq 0, \; x \in \mathbb{R}^3$$

(a) Determine a parametric representation of the central path. That is, find expressions for each coordinate as functions of $\mu$.

Solution. The central path is defined as the locus of minimizers of the Barrier function

$$B_\mu(x) = x_1 + x_3 - \mu \log x_1 - \mu \log x_2 - \mu \log x_3 \quad \text{with } x_2 = 1 - x_1 - x_3.$$

The minimizers are given by the stationarity condition $\nabla_x B_\mu(x) = 0$. as $B_\mu$ is a convex function on the feasible domain.

$$B_\mu(x_1, x_3) = x_1 + x_3 - \mu \log x_1 - \mu \log x_3 - \mu \log(1 - x_1 - x_3)$$

$$\left\{ \begin{aligned} \frac{\partial B_\mu}{\partial x_1} &= 1 - \frac{\mu}{x_1} + \frac{\mu}{1 - x_1 - x_3} \overset{\text{set}}{=} 0 \\ \frac{\partial B_\mu}{\partial x_3} &= 1 - \frac{\mu}{x_3} + \frac{\mu}{1 - x_1 - x_3} \overset{\text{set}}{=} 0 \end{aligned} \right\} \Rightarrow \frac{\mu}{x_1} = \frac{\mu}{x_3} \Rightarrow x_1 = x_3$$

And, $1 - \frac{\mu}{x_1} + \frac{\mu}{1 - 2x_1} = 0 \Rightarrow (1 - 2x_1)x_1 - \mu(1 - 2x_1) + \mu x_1 = 0$

$$\Rightarrow 2x_1^2 - x_1 + \mu - 2\mu x_1 - \mu x_1 = 0$$

$$\Rightarrow 2x_1^2 - (1 + 3\mu)x_1 + \mu = 0$$

$$\Rightarrow x_1 = \frac{(1 + 3\mu) - \sqrt{(1 + 3\mu)^2 - 8\mu}}{4}$$

where we take the negative root to satisfy the constraints.

The result is

$$\boxed{\begin{aligned} x_1(\mu) = x_3(\mu) &= \frac{(1+3\mu)}{4} - \frac{1}{4}\sqrt{(1+3\mu)^2 - 8\mu} \\ x_2(\mu) &= 1 - 2x_1(\mu) \end{aligned}}$$

(b) Show that as $\mu \to 0$, $x(\mu)$ converges to the unique optimal solution.

Solution. We have, for small $\mu$:

$$x_1(\mu) = \frac{1+3\mu}{4} - \frac{1}{4}\sqrt{1+9\mu^2-2\mu} = \frac{1+3\mu}{4} - \frac{1}{4}(1-\mu) + O(\mu^2)$$

$$= \mu + O(\mu^2)$$

So $\lim_{\mu \to 0} x_1(\mu) = 0$ and $\boxed{\lim_{\mu \to 0} x(\mu) = (0,0,1)}$

By inspection, this is the optimal solution, with $f^* = 0$.

(c) Find the analytic center of the feasible region by carefully determining $\lim_{\mu\to\infty} x(\mu)$.

Solution. Let $\delta = \frac{1}{\mu}$ and determine $\lim_{\delta\to 0} x(\delta)$.

$$x_1(\delta) = \frac{1+3/\delta}{4} - \frac{1}{4}\sqrt{1+9/\delta^2-2/\delta} = \frac{3+\delta}{4\delta} - \frac{1}{4\delta}\sqrt{\delta^2-2\delta+9}$$

$$= \frac{1}{4}+\frac{3}{4\delta}-\frac{3}{4\delta}\sqrt{1-\frac{2}{9}\delta+\frac{1}{9}\delta^2} = \frac{1}{4}+\frac{3}{4\delta}-\frac{3}{4\delta}\left(1-\frac{2}{9}\delta+\frac{1}{9}\delta^2\right)^{1/2}$$

$$= \frac{1}{4}+\frac{3}{4\delta}-\frac{3}{4\delta}\left(1-\frac{1}{9}\delta+\frac{1}{18}\delta^2-\frac{1}{162}\delta^2\right)+O(\delta^2) = \frac{1}{4}+\frac{1}{12}-\frac{\delta}{27}+O(\delta^2)$$

$$= \frac{1}{3}-\frac{\delta}{27}+O(\delta^2).$$

So, the analytic center is $\boxed{\lim_{\delta\to 0} x(\mu) = \left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)}$